Type MS · Angle between a line and a plane

Answer 1 of 5 · Question 4 part whole · 6 marks

2024 NOV · Paper 01 · find part whole in this mark scheme extract

4	$(FD)=\sqrt{15^2+20^2}$ $\quad(FD=)25$ $(BF=)15\sin 40$ oe $\quad(\Rightarrow (BF)=9.641814145)$ $\sin BDF=\dfrac{"15\sin 40"}{"\sqrt{15^2+20^2}"}(=0.3856725658)$ or $\sin^{-1}\left(\dfrac{"9.641814145"}{"25"}\right)$ $(BDF)=22.7°$	M1 A1 M1 M1 dM1 A1 [6]
Example of an ALT	$(DC=)15\times\cos 40$ and $(BD=)\sqrt{20^2+("15\cos 40")^2}$ $\quad(BD=)$awrt 23.1 $(DC=)15\times\cos 40$ and $(FB=)\sqrt{15^2-(15\cos 40)^2}$ oe $\quad(\Rightarrow (FB)=9.641814145)$ $\tan BDF=\dfrac{"\sqrt{15^2-(15\cos 40)^2}"}{"\sqrt{20^2+("15\cos 40")^2}"}(=0.4180117433)$ or $\tan^{-1}\left(\dfrac{"9.641814145"}{"23.1"}\right)$ $(BDF)=22.7°$	M1 A1 M1 dM1 M1 A1 [6]
		Total 6 marks

Type MS · Angle between a line and a plane

Answer 2 of 5 · Question 3 part b · 3 marks

2020 NOV · Paper 01 · extract 1 of 20 · find part b in this mark scheme extract

3(a)	$AM = \sqrt{6^2+8^2} = 10$	M1
	$AE = \sqrt{14^2+10^2} = \sqrt{296} = 17.20... = 17.2\,\text{cm}$	M1A1 (3)
(b)	$\tan\phi = \frac{EM}{MA} = \frac{14}{10}$, $\phi = 54.46... = 54.5°$ or using another trig function	M1A1ft,A1(3)
(c)	$\tan\theta = \frac{EM}{\frac{1}{2}CD} = \frac{14}{8}$, $\theta = 60.255...° = 60.3°$	M1A1ft,A1 (3) [9]

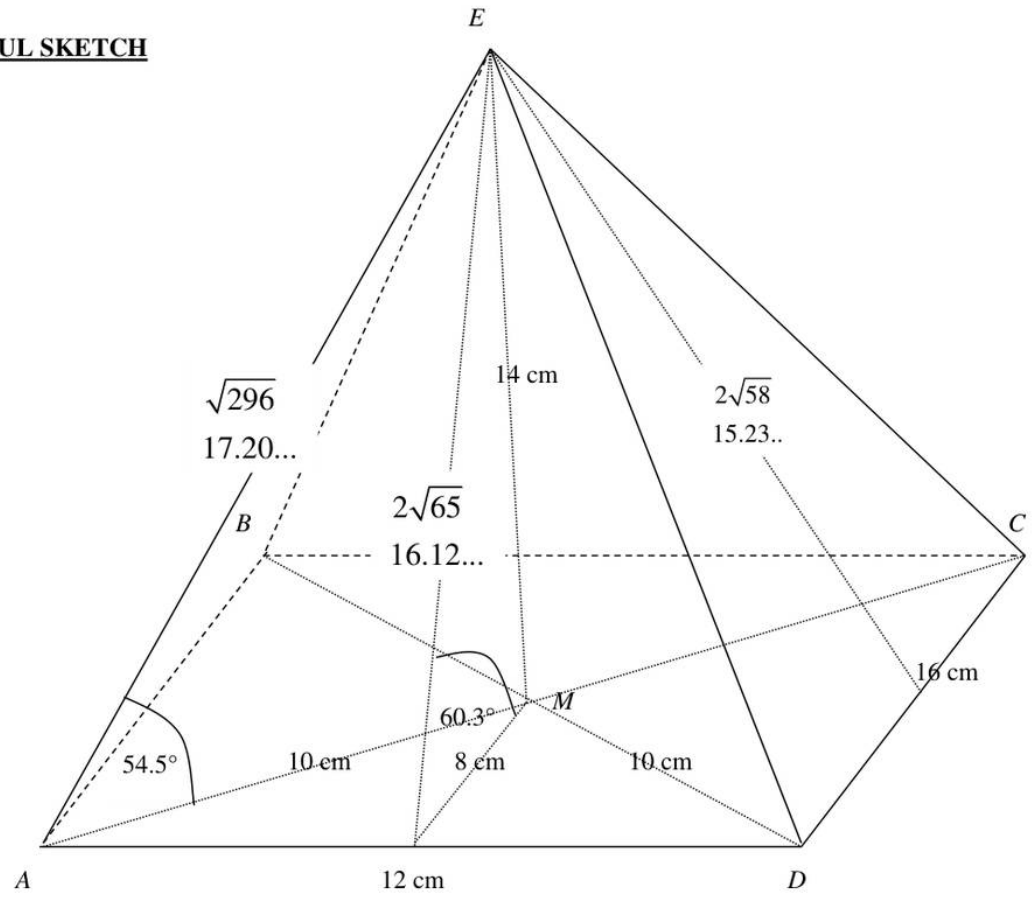
Part	Mark	Notes
(a)	M1	Applies Pythagoras theorem to find the length of AM $AM = \sqrt{6^2+8^2} = 10$ or $AM = \frac{\sqrt{12^2+16^2}}{2} = 10$
	M1	Applies Pythagoras to find the length of one of the sloping edges $AE = \sqrt{14^2+10^2} = \sqrt{296} = ...$
	A1	For the correct length of either AE, DE, CE or BE $AE = 17.2\,\text{cm}$ rounded correctly
	ALT	
	M1M1	Applies Pythagoras in 3D $AE = \sqrt{14^2+6^2+8^2} = \sqrt{296} = ...$
(b)	M1	For applying any acceptable trigonometry to find the required angle. $\tan\phi = \frac{EM}{MA} = \frac{14}{10}$, or $\sin\phi = \frac{14}{\sqrt{296}}$, or $\cos\phi = \frac{10}{\sqrt{296}} \Rightarrow \phi = ...$
	A1ft	For the correct trigonometry if they use sine or cosine following through their $\sqrt{296}$
	A1	Required angle = 54.5° Rounded correctly
(c)	M1	For applying trigonometry to find the required angle. $\tan\theta = \frac{EM}{\frac{1}{2}CD} = \frac{14}{8} \Rightarrow \theta = ...$ OR The length of the perpendicular from E to the mid-point of AD is $\sqrt{260}$ $\sin\theta = \left(\frac{14}{'\sqrt{260}'}\right)$, or $\cos\left(\frac{8}{'\sqrt{260}'}\right) \Rightarrow \theta = ...$
	A1ft	Ft their $\sqrt{260}$
	A1	$\theta = 60.3°$
Rounding: Penalise rounding only the first time it occurs in either (b) or (c)		

Type MS · Angle between a line and a plane

Answer 2 of 5 · Question 3 part b · 3 marks

2020 NOV · Paper 01 · extract 2 of 20 · find part b in this mark scheme extract

USEFUL SKETCH



Type MS · Angle between a line and a plane

Answer 2 of 5 · Question 3 part b · 3 marks

2020 NOV · Paper 01 · extract 3 of 20 · find part b in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Angle between a line and a plane

Answer 2 of 5 · Question 3 part b · 3 marks

2020 NOV · Paper 01 · extract 4 of 20 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	Divides through $x^3-4x^2+5=0$ by x^2 and rearranges to achieve as a minimum $x+\frac{5}{x^2}=k$ where k is a constant $\left[x-4+\frac{5}{x^2}=0\Rightarrow x+\frac{5}{x^2}=4\right]$
	M1	**Draw the line $y=k$** following through their value for k No line is M0
	A1	For the two values of $x=1.4$ and $x=3.6$ Condone answers given as coordinates provided they are completely correct. (1.4, 4) and (3.6, 4) Require both M marks for this mark.
(b)	M1	For setting $x+\frac{5}{x^2}=Ax+B\Rightarrow x^3+5=Ax^3+Bx^2\Rightarrow Ax^3-x^3+Bx^2-5=0$, and equating coefficients with x^3-x^2-5 $x^3(A-1)+Bx^2-5\equiv x^3-x^2-5$ to achieve as a minimum $A=(\pm 2)$, $B=(\pm 1)$
	A1	For the correct straight line $y=2x-1$
	ALT – to find the line $y=2x-1$	
	M1	Divides through $x^3-x^2-5=0$ by x^2 and rearranges the equation to achieve as a minimum $\Rightarrow \pm 2x\pm 1=x+\frac{5}{x^2}$
	A1	For the correct straight line $y=2x-1$
	dM1	**Draws their $y=2x-1$ on the graph** and locates the point of intersection. Please check that they draw their line correctly. Coordinates for you to check are (0.5, 0) and (2.5, 4) No line is M0 This mark is dependent on the first M mark in (b)
	A1	For the correct value of $x=2.1$ [allow $x=2.2$] Can only score this mark from M1A1M1 Do not accept the answer given as coordinates.

Type MS · Angle between a line and a plane

Answer 2 of 5 · Question 3 part b · 3 marks

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Question Number	Scheme	Marks
5(a)	Gradient $PR = \frac{6}{-6} = -1$, Gradient $QS = \frac{8}{8} = 1$	M1A1
	Product $= -1 \Rightarrow$ perpendicular	A1 (3)
(b)	(i) $PR = \sqrt{6^2 + 6^2} = 6\sqrt{2}$ $\left(= \sqrt{72}\right)$	M1
	(ii) $QS = \sqrt{8^2 + 8^2} = 8\sqrt{2}$ $\left(= \sqrt{128}\right)$	A1 (2)
(c)	Area $= \frac{1}{2}$ "$6\sqrt{2}$"×"$8\sqrt{2}$" $= 48$ $\left(\text{units}^2\right)$	M1A1 (2)
		[7]

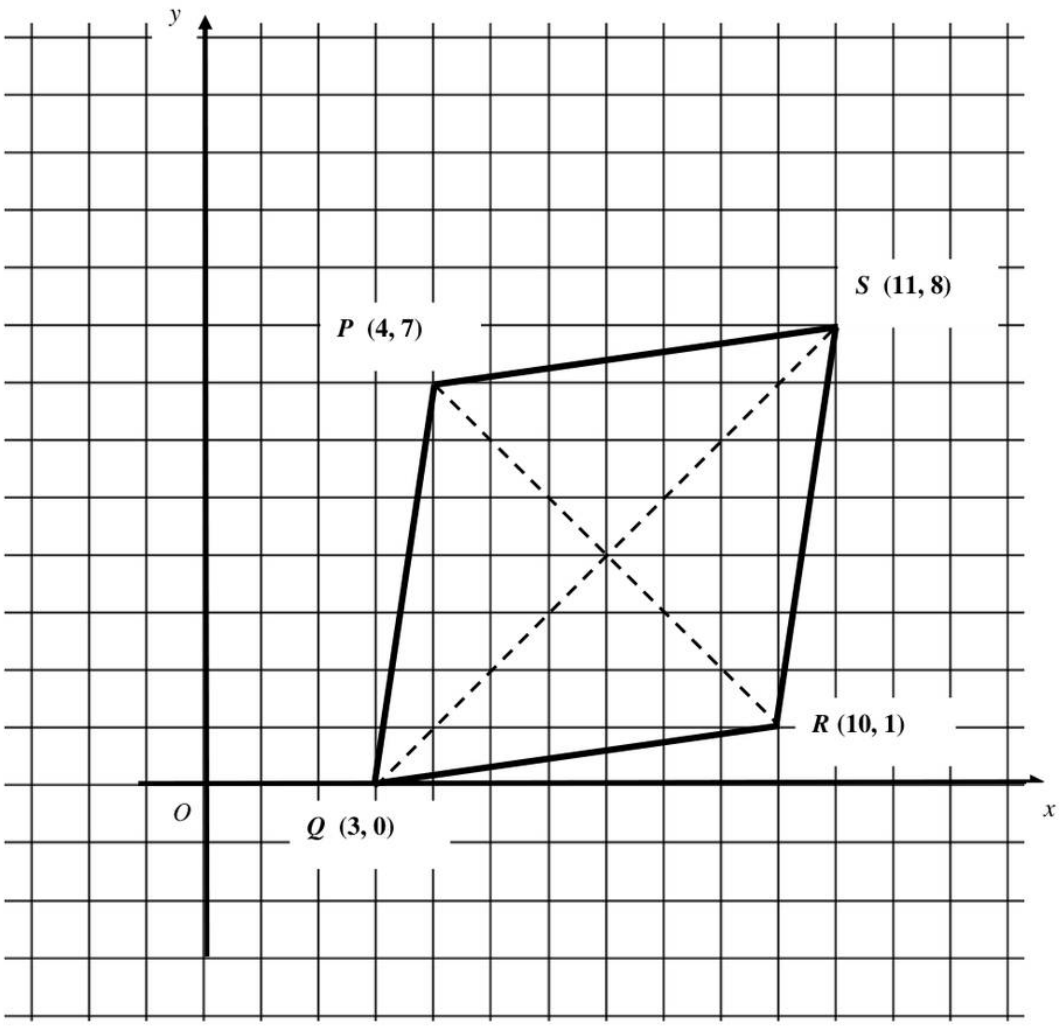
Part	Mark	Notes
(a)	M1	Finds the gradient of PR and QS using a correct method. This may be on a diagram. Gradient $PR = \frac{7-1}{4-10} = \frac{6}{-6} = -1$, Gradient $QS = \frac{8-0}{11-3} = \frac{8}{8} = 1$
	A1	Both gradients correct Gradient $PR = -1$, Gradient $QS = 1$
	A1	Finds the product of the two gradients with a statements that as the product = –1 then the lines are perpendicular.
(b)	M1	For **either** $PR = \sqrt{6^2 + 6^2} = \sqrt{72}$ or $6\sqrt{2}$ **OR** $QS = \sqrt{8^2 + 8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
	A1	For **both** $PR = \sqrt{6^2 + 6^2} = \sqrt{72}$ or $6\sqrt{2}$ **AND** $QS = \sqrt{8^2 + 8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
(c)	M1	$PQRS$ is a kite so $= \frac{1}{2}$ "$6\sqrt{2}$"×"$8\sqrt{2}$" $= \ldots$
	A1	Area = 48 $\left(\text{units}^2\right)$
	ALT Uses determinants	
	M1	Area $= \frac{1}{2}\begin{pmatrix} 4 & 3 & 10 & 11 & 4 \\ 7 & 0 & 1 & 8 & 7 \end{pmatrix}$ $= \frac{1}{2}\left(\left[4\times0+3\times1+10\times8+11\times7\right]-\left[3\times7+10\times0+11\times1+4\times8\right]\right)$ $= \ldots$ Allow one slip in a product.
	A1	Area = 48 $\left(\text{units}^2\right)$

Type MS · Angle between a line and a plane

Answer 2 of 5 · Question 3 part b · 3 marks

2020 NOV · Paper 01 · extract 6 of 20 · find part b in this mark scheme extract

USEFUL SKETCH



Type MS · Angle between a line and a plane

Answer 2 of 5 · Question 3 part b · 3 marks

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Type MS · Angle between a line and a plane

Answer 2 of 5 · Question 3 part b · 3 marks

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Question Number	Scheme	Marks
6(a)	$a = S_1 = 1(15+2\times1) = 17$	B1
	$S_2 = 2(15+2\times2)(=38) = 2a+d$	M1A1
	$2\times17+d=38 \Rightarrow d=4$	A1 (4)
(b)	20th term $= a+19d = 17+19\times4 = 93$	M1A1 (2)
(c)	$S_{2p}-2S_p = 1+S_{p-1}$	
	$2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$	M1
	$2p^2-11p+12=0$	A1
	$(2p-3)(p-4)=0 \Rightarrow p=4$ $\left(p\neq\frac{3}{2};\text{ may not be seen}\right)$	M1A1 (4)
		[10]

Part	Mark	Notes
(a)	B1	For the first term $a = 17$ $[a = S_1 = 1(15+2\times1) = 17]$
	M1	For the second term. Uses the given summation formula to form a linear equation in a and d for a minimally acceptable response of $k = 2a+d$ where k is a positive integer.
	A1	For the correct linear equation $38 = 2a+d$
	A1	For the correct value of $d = 4$
	ALT 1	
	B1	For the first term $a = 17$
	M1	For using a correct summation formula $n(15+2n) = \frac{n}{2}(2a+[n-1]d) \Rightarrow 30+2n = 2a-d+nd$ and equates coefficients
	A1	For equating coefficients of n $4n = dn \Rightarrow d = \ldots$ and $30 = 2a-4 \Rightarrow a = \ldots$
		For the correct value of $d = 4$

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Answer 2 of 5 · Question 3 part b · 3 marks

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	ALT 2	
	B1	For the first term $a = 17$
	M1	Uses two values of n to set up a pair of simultaneous equations.e.g. $S_4 = 4(15+2\times4) = 92$ and $92 = \frac{4}{2}(2a+3d) \Rightarrow 46 = 2a+3d$ $S_5 = 5(15+2\times5) = 125$ and $125 = \frac{5}{2}(2a+4d) \Rightarrow 50 = 2a+4d$
	A1	Attempts to solve the pair of equations
	A1	$d = 4$
(b)	M1	For using the correct nth term formula with **their a and their d** $U_{20} = '17'+19\times'4' = ...$
	A1	For the correct 20th term = 93
(c)	M1	Uses the given summation formula with the correct substitution $2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	ALT	
	M1	Uses the summation formula: **Follow through their a and d** $S_{2p} = \frac{2p}{2}(2\times17+(2p-1)4) = p(30+8p)$ $2S_p = 2\times\frac{p}{2}(2\times17+(p-1)4) = p(30+4p)$ $S_{p-1} = \frac{p-1}{2}(2\times17+(2[p-1]-1)4) = (p-1)(13+2p)$ For a correct substitution into the given expression $p(30+8p)-p(30+4p) = 1+(p-1)(13+2p)$ oe
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	M1	For attempting to solve their 3TQ (provided it is a 3TQ) by any valid method. $2p^2-11p+12 = (2p-3)(p-4) = 0 \Rightarrow p = ..., ...$
	A1	For $p = 4$ If they give both roots of their 3TQ as an answer without rejecting $p = 1.5$ A0

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Answer 2 of 5 · Question 3 part b · 3 marks

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Question Number	Scheme	Marks
7(a)	$x^2-9x+14=\left(x-\frac{9}{2}\right)^2+14-\frac{81}{4}=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$	M1
	$a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe	A1 (2)
(b)	(i) least value of $\mathrm{f}(x)=-\frac{25}{4}$	B1ft
	(ii) least value when $x=\frac{9}{2}$	B1ft (2)
(c)	$x+5=x^2-9x+14$	M1
	$x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$	M1
	Points are $(9,14)$ $(1,6)$	A1A1 (4)
(d)	Area $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)\mathrm{d}x=\int_1^9\left(-x^2+10x-9\right)\mathrm{d}x$	M1
	$=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$	M1A1
	$=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=85\frac{1}{3}$	M1A1 (5)
		[13]

Part	Mark	Notes
(a)	M1	For attempting to complete the square to achieve as a minimum $x^2-9x+14=\left(x\pm\frac{9}{2}\right)^2+14-k$ where k is a constant
	A1	For the correct expression $x^2-9x+14=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$ or $a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe stated
(b)(i)	B1ft	For the correct value $\mathrm{f}(x)=-\frac{25}{4}$ follow through their value of '$-\frac{25}{4}$'
(ii)	B1ft	For the correct value of $x=\frac{9}{2}$ provided they have $\left(x-\frac{9}{2}\right)^2$ …in part (a). Follow through their value of '$\frac{9}{2}$'

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Answer 2 of 5 · Question 3 part b · 3 marks

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(c)	M1	For equating the equation of the line with the equation of C $x+5=x^2-9x+14\Rightarrow x^2-10x+9=0$ **and** attempting to form a 3TQ
	M1	Attempts to solve their 3TQ by any method, provided it is the result of equating the line with C $x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$
	A1	For the correct coordinates of **either** $(9,14)$ or $(1,6)$
	A1	For both correct pairs of coordinates $(9,14)$ and $(1,6)$
(d)	M1	For a correct expression for the required area with both limits correct. (ft their limits from (c)) Award this mark if they have 'curve – line' but otherwise correct. $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)\mathrm{d}x=\left[\int_1^9\left(-x^2+10x-9\right)\mathrm{d}x\right]$, accept $\int_1^9\left(x^2-10x+9\right)\mathrm{d}x$ **OR** Area under the trapezium – curve $\frac{1}{2}\times8\times(6+14)-\int_1^9\left(x^2-9x+14\right)\mathrm{d}x$
	M1	For attempting to integrate the equation for the combined expression or the curve only.
	A1	For the correct integrated expression for required area. Ignore limits for this mark – even if they are absent altogether. Area = $=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$ accept $\left[\frac{x^3}{3}-5x^2+9x\right]_1^9$ **OR** $\frac{1}{2}\times8\times(6+14)-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ **OR** $\left(\frac{x^2}{2}+5x\right)_1^9-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ or $\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9-\left(\frac{x^2}{2}+5x\right)_1^9$
	M1	For substituting their limits (x – coordinates from part(c)) into their integrated expression. $=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=\ldots$ **OR** $80-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$ **OR** $\left[\left(\frac{9^2}{2}+5\times9\right)-\left(\frac{1^2}{2}+5\times1\right)\right]-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$
	A1	For the correct area of $85\frac{1}{3}$ or $\frac{256}{3}$ If they get a value of $-85\frac{1}{3}$ they must give a **final** value of $85\frac{1}{3}$ for this mark.

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Answer 2 of 5 · Question 3 part b · 3 marks

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Question Number	Scheme	Marks
8(a)	$2xy+5y=e^x \quad y=\dfrac{e^x}{(2x+5)}$	
	$\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$	M1A1A1
	$\dfrac{dy}{dx}=\dfrac{e^x}{(2x+5)}\times\dfrac{(2x+5-2)}{(2x+5)}=\dfrac{y(2x+3)}{(2x+5)}$ *	M1A1 (5)
(b)	$x=0\Rightarrow \dfrac{dy}{dx}=\dfrac{5-2}{5^2}=\dfrac{3}{25}$	M1A1 (2)
ALT	$x=0\Rightarrow y=\dfrac{1}{5},\ \dfrac{dy}{dx}=\dfrac{1}{5}\times\dfrac{3}{5}=\dfrac{3}{25}$	
(c)	$x=0\Rightarrow y=\dfrac{e^0}{(2\times0+5)}=\dfrac{1}{5}$	M1(Award if seen in (b) and **used** in (c))
	$y-\dfrac{1}{5}=-\dfrac{25}{3}x$	M1
	$125x+15y-3=0$	A1 (3) [10]

Part	Mark	Notes
(a)		$2xy+5y=e^x \Rightarrow y=\dfrac{e^x}{(2x+5)}$
	M1	For attempting Quotient Rule • Both terms must be differentiated correctly $e^x\Rightarrow e^x \quad 2x+5\Rightarrow 2$ • There must be two terms subtracted in the numerator either way around • The denominator must the denominator squared. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$
	A1	For $e^x(2x+5)$ or $2e^x$
	A1	For a fully correct differentiated expression. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$

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	M1	Subs in $y=\frac{e^x}{(2x+5)}$ as a common factor $\frac{dy}{dx}=\frac{e^x}{(2x+5)}\times\frac{(2x+5-2)}{(2x+5)}=\frac{y(2x+5-2)}{(2x+5)}$	Subs in $e^x=y(2x+5)$ and factorises $\frac{dy}{dx}=\frac{y(2x+5)(2x+5)-2y(2x+5)}{(2x+5)^2}$
	A1	For the correct answer with no errors. **Note this is a given answer.** $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
	ALT – uses implicit differentiation on $2xy+5y=e^x$		
	M1	$2\left(y+x\frac{dy}{dx}\right)+5\frac{dy}{dx}=e^x$	
	A1	Takes out $\frac{dy}{dx}$ as a common factor $\frac{dy}{dx}(2x+5)=e^x-2y$	
	A1	For a fully correct differentiated expression as below. $\frac{dy}{dx}=\frac{e^x-2y}{(2x+5)}$	
	M1	For separating the fraction, taking out y as a common factor and attempting to form a single fraction $\frac{dy}{dx}=\frac{e^x}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5)}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5-2)}{2x+5}$	
	A1	For the correct answer with no errors. $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
(b)	M1	For substituting $x=0$ into $\frac{dy}{dx}=\frac{e^x(2x+5)-2e^x}{(2x+5)^2}=\frac{e^0(2\times0+5)-2e^0}{(2\times0+5)^2}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
	ALT		
	M1	When $x=0\Rightarrow y=\frac{1}{5}$, $\frac{dy}{dx}=\frac{1}{5}\times\frac{3}{5}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
(c)	M1	$x=0\Rightarrow y=\frac{e^0}{(2\times0+5)}=\frac{1}{5}$ $x=0\Rightarrow y=\frac{1}{5}$ Award if seen in (b) and **used** in (c) **This is a B mark in Epen.**	
	M1	Inverts the gradient found in (b) and forms equation of the normal. ft their value of y $y-\frac{1}{5}=-\frac{25}{3}x$	
	A1	Equation of line is given in the required form. $125x+15y-3=0$	

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Answer 2 of 5 · Question 3 part b · 3 marks

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Question Number	Scheme	Marks
9(a)	$\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$	M1A1 (2)
(b)	$\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$	B1
	$ADB = 180 - BDC \Rightarrow -\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$	M1A1
	$2x^2 = 108 \Rightarrow x = 3\sqrt{6}$	
	$AC = 6\sqrt{6}$	A1 (4)
(c)	$\dfrac{\sin(\theta^\circ + \phi^\circ)}{2x} = \dfrac{\sin BCD}{12} \Rightarrow \dfrac{\sin(\theta^\circ + \phi^\circ)}{x} = \dfrac{\sin BCD}{6}$	M1A1
	$\dfrac{\sin\phi^\circ}{x} = \dfrac{\sin BCD}{6}$	M1
	$\therefore \sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$	A1 (4)
(d)	$\sin(\theta^\circ + \phi^\circ) = \sin\phi^\circ \Rightarrow (\theta + \phi) = 180 - \phi$ (or ϕ or $360 + \phi$ (not possible))	M1
	$\therefore \theta = 180 - 2\phi$	A1 (2) [12]

Part	Mark	Notes
(a)	M1	For using a correct cosine rule $\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$ or $12^2 = x^2 + 6^2 - 2\times 6\times x\cos ADB$
	A1	Simplifies to $\cos ADB = \dfrac{x^2 - 108}{12x}$
(b)	B1	For the correct expression $\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$
	M1	$\cos BDC = -\cos ADB$ and $\cos BDC = -\dfrac{x^2}{12x}$ so $-\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$ and attempts to solve
	ALT 1 – uses triangles *BAD* and *BAC*	
	B1	For **both** of the following correct expressions for $\cos BAD$ and $\cos BAC$: $\cos BAD = \dfrac{12^2 + x^2 - 6^2}{2\times 12\times x}$ and $\cos BAC = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x}$
	M1	$\angle BAC = \angle BAD$ so equates their two expressions $\dfrac{12^2 + x^2 - 6^2}{2\times 12\times x} = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x} \Rightarrow \dfrac{108 + x^2}{24x} = \dfrac{108 + 4x^2}{48x}$ and attempts to solve
	ALT 2 – uses triangles *BCD* and *BCA*	

Type MS · Angle between a line and a plane

Answer 2 of 5 · Question 3 part b · 3 marks

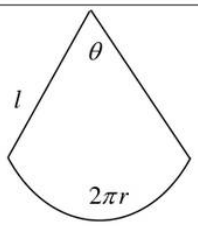
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	B1	For **both** of the following correct expressions for cos BAD and cos BAC: $\cos BCD = \frac{6^2 + x^2 - 6^2}{2\times 6\times x}$ and $\cos BCA = \frac{6^2 + (2x)^2 - 12^2}{2\times 6\times 2x}$
	M1	$\frac{6^2 + x^2 - 6^2}{2\times 6\times x} = \frac{6^2 + (2x)^2 - 12^2}{2\times 6\times 2x} \Rightarrow x^2 = 2x^2 - 54$ and attempts to solve
	Final A marks for all three methods	
	A1	For the correct value of $x = 3\sqrt{6}$
	A1	For $AC = 6\sqrt{6}$
(c)	M1	Uses sine rule on triangle ABC: $\frac{\sin(\theta^\circ + \phi^\circ)}{2x} = \frac{\sin BCD}{12} \Rightarrow \frac{\sin(\theta^\circ + \phi^\circ)}{x} = \frac{\sin BCD}{6}$
	A1	Achieves the correct expression for $\sin(\theta^\circ + \phi^\circ) = \frac{x\sin BCD}{6}$
	M1	Uses sine rule on triangle BDC: $\frac{\sin\phi^\circ}{x} = \frac{\sin BCD}{6} \Rightarrow \left(\sin\phi^\circ = \frac{x\sin BCD}{6}\right)$
	A1	Shows that $\sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$ with no errors
	ALT 1 – Uses exact values for the trigonometric ratios and the expansion for sin $(A + B)$	
	M1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **or** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **or** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	A1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **and** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **and** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	M1	Expands $\sin(\theta + \phi) = \frac{\sqrt{15}}{8}\times\frac{1}{4} + \frac{\sqrt{15}}{4}\times\frac{7}{8} = \left(\frac{\sqrt{15}}{4}\right)$ Or $\sin(\theta + \phi)^\circ = \sin(28.95^\circ)\cos(75.52^\circ) + \sin(75.52^\circ)\cos(28.95^\circ) = 0.968... = \sin\phi^\circ$
	A1	Shows that $\sin(\theta + \phi) = \frac{\sqrt{15}}{4}$ and $\sin\phi = \frac{\sqrt{15}}{4}$ so $\sin(\theta + \phi) = \sin\phi$ with no errors. If they use approximate values for sin θ and ϕ withhold this final mark so A0
	ALT 2 – Uses $\angle BCD = 52.2...^\circ$	
	M1	Finds $\angle BCD = 52.2...^\circ$ using cosine rule and applies sine rule on triangle ABC $\frac{\sin(\theta^\circ + \phi^\circ)}{6\sqrt{6}} = \frac{\sin 52.2^\circ}{12}$
	A1	Shows that $\sin(\theta^\circ + \phi^\circ) = 0.968...$
	M1	Uses sine rule on triangle BD: $\frac{\sin\phi^\circ}{3\sqrt{6}} = \frac{\sin 52.2^\circ}{6} \Rightarrow \sin\phi^\circ = 0.968... = \sin(\theta^\circ + \phi^\circ)$
	A1	If they use an approximate value for angle BCD withhold this final mark so A0
(d)	M1	For writing $\sin\phi^\circ = \sin(180 - \phi)^\circ \Rightarrow \theta^\circ + \phi^\circ = 180^\circ - \phi^\circ$
	A1	For rearranging $\theta^\circ + \phi^\circ = 180^\circ - \phi^\circ$ to achieve $\therefore \theta = 180 - 2\phi$ This is a show question and there must be no errors here.

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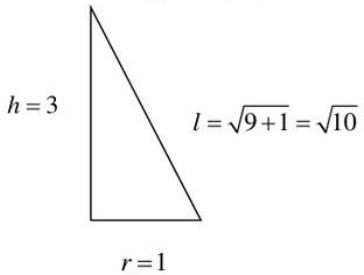
Question Number	Scheme	Marks
10(a)	Circumference of base $= 2\pi r$ 	B1
	$l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	B1
	$A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l} = \pi rl$	M1A1 (4)
(b)	$A = \pi rl$	
	$l = r\sqrt{10} \Rightarrow A = \pi r^2\sqrt{10}$	B1
	$\dfrac{\mathrm{d}A}{\mathrm{d}r} = 2\pi r\sqrt{10} \Rightarrow k = 2\sqrt{10}$	M1A1 (3)
(c)	$\dfrac{\mathrm{d}V}{\mathrm{d}t} = 1.5\ \left(\mathrm{cm}^3/\mathrm{s}\right)$	B1
	$\dfrac{\mathrm{d}A}{\mathrm{d}t} = \dfrac{\mathrm{d}A}{\mathrm{d}r} \times \dfrac{\mathrm{d}r}{\mathrm{d}V} \times \dfrac{\mathrm{d}V}{\mathrm{d}t}$	M1
	$V = \dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi r^2 \times 3r = \pi r^3$	B1
	$\dfrac{\mathrm{d}V}{\mathrm{d}r} = 3\pi r^2$	B1ft
	$\therefore \dfrac{\mathrm{d}A}{\mathrm{d}t} = 2\pi \times 8\sqrt{10} \times 1.5 \times \dfrac{1}{3 \times 64\pi} = 0.3952... = 0.395\ \mathrm{cm}^2/\mathrm{s}$	A1 (5)
		[12]

Part	Mark	Notes	
(a)	B1	For the circ of the base $L = 2\pi r$	
	B1	$R = l$ and $l = r\theta$ Therefore $l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	$R = l$
	M1	$A = \dfrac{1}{2}l^2\theta$ and substituting their expression for θ to give $A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l}$	Uses the formula $A = \dfrac{1}{2}RL$ $A = \dfrac{1}{2} \times l \times 2\pi r \Rightarrow \left(A = \pi rl\right)$
	A1	For the required expression for A, $A = \pi rl$ with no errors.	

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(b)	B1	For finding that the slant height is $\sqrt{10}$ times the radius of the cone  So $l=r\sqrt{10}$
	M1	Substitutes $l=r\sqrt{10}$ into the given expression $A=\pi rl$ and differentiates their resulting expression to find $\frac{dA}{dr}$ $A=\pi r^2\sqrt{10}$ so $\frac{dA}{dr}=2\pi r\sqrt{10}$
	A1	Therefore $k=2\sqrt{10}$
(c)	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dr}\times\frac{dr}{dV}\times\frac{dV}{dt}$
	B1	For finding the volume of a cone in terms of r only $V=\frac{1}{3}\pi r^2h=\frac{1}{3}\pi r^2\times 3r=\pi r^3$
	B1ft	Differentiates their expression for the volume of a cone provided it is in terms of V and r only. $\frac{dV}{dr}=3\pi r^2$
	A1	For combining all required terms into their chain rule and evaluating to 3 significant figures, $\frac{dA}{dt}=2\pi\times 8\sqrt{10}\times 1.5\times\frac{1}{3\times 64\pi}=0.3952...=0.395\ \text{cm}^2/\text{s}$
	ALT – in terms of h	
	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dh}\times\frac{dh}{dt}$ and $\frac{dV}{dt}=\frac{dV}{dh}\times\frac{dh}{dt}$
	B1	For finding the area and volume of a cone in terms of h $r=\frac{h}{3}$, $A=\pi r^2\sqrt{10}\Rightarrow A=\frac{\sqrt{10}}{9}\pi h^2$ and $V=\frac{1}{3}\pi r^2h\Rightarrow V=\frac{1}{27}\pi h^3$
	B1ft	Differentiates their expressions for the area and volume of a cone provided they are both in terms of h only. $\frac{dA}{dh}=\frac{2\sqrt{10}}{9}\pi h$ and $\frac{dV}{dh}=\frac{3}{27}\pi h^2$
	A1	Combines the required terms into their chain rules and evaluating to 3 significant figures $\frac{dA}{dt}=0.395\ 102$

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Question Number	Scheme	Marks
11(a)	(i) $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$	B1
	(ii) $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC} = \mathbf{a} + \frac{1}{2}\left(-\mathbf{a} + \frac{1}{4}\mathbf{b}\right) = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$	M1A1
	(iii) $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD} = -\mathbf{b} + \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$	M1A1 (5)
(b)	$\overrightarrow{BE} = k\overrightarrow{BD} = k\left(\frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}\right)$	M1
	$\overrightarrow{BE} = -\mathbf{b} + \overrightarrow{OE} = -\mathbf{b} + \lambda\mathbf{a}$	M1
	$\frac{7}{8}k = 1 \Rightarrow k = \frac{8}{7}$	M1
	$\frac{k}{2} = \lambda \Rightarrow \lambda = \frac{4}{7}$	A1 (4)
(c)	$\triangle OAC = \frac{1}{4}\triangle OAB$	M1
	$\triangle OEB = "\frac{4}{7}"\triangle OAB$	A1 ft
	$\frac{\triangle OAC}{\triangle OEB} = \frac{1}{4} \times "\frac{7}{4}" = \frac{7}{16}$	M1
	$\mu = \frac{7}{16}$	A1 (4)
		[13]

Part	Mark	Notes
(a)(i)	B1	For the correct vector $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$
(ii)	M1	For the correct vector statement $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC}$ or $\overrightarrow{OD} = \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CA}$
	A1	For the correct **simplified** vector $\overrightarrow{OD} = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$
(iii)	M1	For the correct vector statement $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD}$ or $\overrightarrow{BD} = \overrightarrow{BC} + \overrightarrow{CD}$
	A1	For the correct **simplified** vector $\overrightarrow{BD} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$

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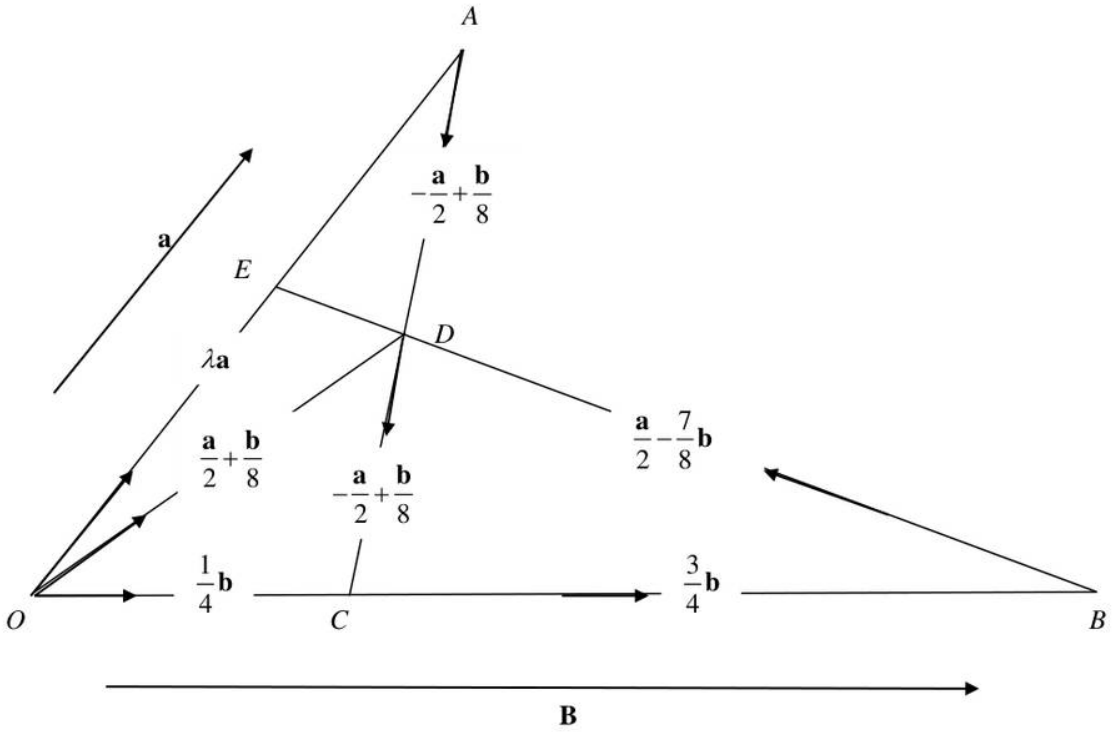
In parts (b) and (c) you must follow through the vectors they have found in part (a)		
(b)	M1	States or implies two paths that will lead to a solution. For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}$ and $\overrightarrow{BE}=\overrightarrow{BO}+\overrightarrow{OE}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\overrightarrow{OD}+\overrightarrow{DE}$ **This is a B mark in Epen**
	M1	For writing their paths as vectors in terms of **a**, **b** λ **and another constant** (e.g. k or μ) For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}=k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ and $\overrightarrow{BE}=-\mathbf{b}+\lambda\mathbf{a}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ OR $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$
	dM1	Equates coefficients of their **two** expressions and attempts to find the value of λ: In triangle OEB • $k\frac{1}{2}\mathbf{a}-k\frac{7}{8}\mathbf{b}=\lambda\mathbf{a}-\mathbf{b}$ $\Rightarrow k\frac{7}{8}=1\Rightarrow k=\frac{8}{7}$ and $\frac{k}{2}=\lambda\Rightarrow\lambda=...$ or in triangle OED, there are two possible expressions for $\overrightarrow{DE}$ • $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ $\Rightarrow k=\frac{1}{8}$ and $\lambda=\frac{1}{2}+\frac{1}{8}\lambda\Rightarrow\lambda=...$ Or $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ $\Rightarrow\frac{7}{8}k=\frac{1}{8}\Rightarrow k=\frac{1}{7}$ and $\lambda=\frac{1}{2}+\frac{1}{7}k\Rightarrow\lambda...$
	A1	For the correct value of λ $\lambda=\frac{4}{7}$
(c)	M1	For stating that $\triangle OAC=\frac{1}{4}\triangle OAB$ or $4\triangle OAC=\triangle OAB$ **1**
	M1	For stating $\triangle OEB="\lambda"\triangle OAB\Rightarrow\triangle OEB="\frac{4}{7}"\triangle OAB$ **2** **This is an A mark in Epen**
	M1	For dividing **1** by **2** $=\frac{\triangle OAC}{\triangle OEB}=\frac{1}{4}\times"\frac{7}{4}"=...$
	A1	For $\mu=\frac{7}{16}$

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USEFUL SKETCH



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5 a	$OC = \frac{1}{2}\sqrt{12^2+12^2} = \left[6\sqrt{2}\right]$ or $AC = \sqrt{12^2+12^2} = \left[12\sqrt{2}\right]$	M1
	$h = EO = 6\sqrt{2} \times \frac{\sqrt{3}}{3} = 2\sqrt{6}$ oe	M1 A1
		(3)
b	Midpoint of AD to $F = \left(6 - \frac{12}{1+4}\right)$ [= 3.6]	M1
	$OF = \sqrt{6^2+3.6^2} = \frac{6\sqrt{34}}{5}$	M1
	$\tan\theta = \frac{2\sqrt{6}}{\frac{6\sqrt{34}}{5}} = 35^\circ$	M1 A1
		(4)
		Total 7 marks

USEFUL SKETCH

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9(a)	$\sin 75^\circ = \sin\left(45^\circ + 30^\circ\right) = \sin 45^\circ \cos 30^\circ + \sin 30^\circ \cos 45^\circ$ $= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{6}+\sqrt{2}}{4} *$ **ALT 1** $\sin 75^\circ = \sin\left(120^\circ - 45^\circ\right) = \sin 120^\circ \cos 45^\circ - \sin 45^\circ \cos 120^\circ$ $= \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \times \left(-\frac{1}{2}\right) = \frac{\sqrt{6}+\sqrt{2}}{4} *$ **ALT 2** $\sin 75^\circ = \sin\left(60^\circ + 15^\circ\right) = \sin 60^\circ \cos 15^\circ + \sin 15^\circ \cos 60^\circ$ $= \frac{\sqrt{3}}{2} \times \left(\frac{\sqrt{6}+\sqrt{2}}{4}\right) + \left(\frac{\sqrt{6}-\sqrt{2}}{4}\right) \times \frac{1}{2}$ $= \frac{3\sqrt{2}+\sqrt{6}}{8} + \frac{\sqrt{6}-\sqrt{2}}{8} = \frac{\sqrt{6}+\sqrt{2}}{4}$	M1 M1A1 cso [3] [M1 M1A1] [M1 M1 A1]
(b)	$\left[\angle AFC = 180^\circ - 60^\circ - 45^\circ = 75^\circ\right]$ $\frac{24}{\sin 75^\circ} = \frac{BF}{\sin 60^\circ}$ $\Rightarrow \frac{24}{\sin 75^\circ} = \frac{BF}{\sin 60^\circ} \Rightarrow \frac{24}{\frac{\sqrt{6}+\sqrt{2}}{4}} = \frac{BF}{\frac{\sqrt{3}}{2}}, \Rightarrow BF = \frac{48\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ $BF = \frac{48\sqrt{3}}{\left(\sqrt{6}+\sqrt{2}\right)} \times \frac{\left(\sqrt{6}-\sqrt{2}\right)}{\left(\sqrt{6}-\sqrt{2}\right)} = \frac{48\sqrt{3}\sqrt{6}-48\sqrt{6}}{6-2} = 12\left(3\sqrt{2}-\sqrt{6}\right) *$	M1,M1A1 M1A1 cso [5]
(c)	Let the vertical point from F to the line AB be X and the corresponding point on CD be Y. The required angle is EXY $FX = 12\left(3\sqrt{2}-\sqrt{6}\right)\sin 45^\circ = 36 - 12\sqrt{3} = (15.2153...)$ OR Area $ABC = \frac{1}{2} \times 24 \times FX = \frac{1}{2} \times 24 \times 12\left(3\sqrt{2}-\sqrt{6}\right)\sin 45$ $\Rightarrow FX = 12\left(3\sqrt{2}-\sqrt{6}\right)\sin 45^\circ = 36 - 12\sqrt{3} = (15.2153...)$ $EF = \frac{36-12\sqrt{3}}{\tan 65^\circ} = 7.09505.... \approx 7.1 \text{ (cm)}$	M1 M1A1 [3]
(d)	Identifies the required angle FCX $FC = \sqrt{\left[12\left(3\sqrt{2}-\sqrt{6}\right)\right]^2 + '7.09505'^2} = 22.65735.....$ $\angle FCX = \sin^{-1}\left(\frac{15.2153}{'22.6573'}\right) = 42.186...^\circ \approx 42.2^\circ$	B1 M1 M1A1 [4]
		Total 15 marks

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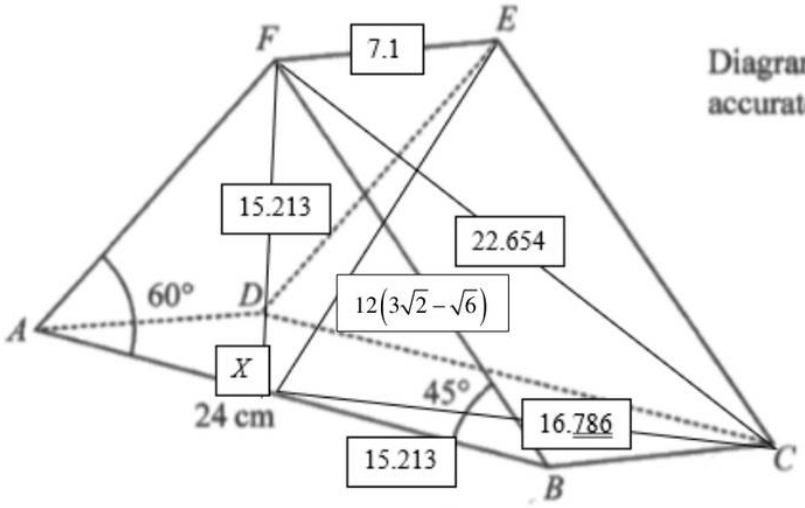
Part	Mark	Notes
(a)	M1	For the correct use of the addition formula for sine with $\sin(45^\circ+30^\circ)$ Accept also $\sin(120^\circ-45^\circ)$ and $\sin(60^\circ+15^\circ)$ **but not** $\sin(90^\circ-15^\circ)$ or $\sin(180^\circ-105^\circ)$
	M1	For applying the exact values of these sines and cosines and for evaluating the expression.. This step must be explicitly seen.
	A1 cso	For the correct final answer with no errors seen
(b)	M1	For the correct use of sine rule using the exact value for sin 75° and sin 60°
	M1	For rearranging to obtain $BF=\frac{k\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ or $\frac{l\sqrt{3}}{2\sqrt{6}+2\sqrt{2}}$ Where k and l are integers
	A1	For the correct **exact** BF which need not be simplified. $BF=\frac{48\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ or $\frac{96\sqrt{3}}{2\sqrt{6}+2\sqrt{2}}$ or even $\frac{12\sqrt{3}\times4}{\sqrt{6}+\sqrt{2}}$ but not for example, $\frac{\frac{12\sqrt{3}}{\sqrt{6}+\sqrt{2}}}{4}$ unless recovery is seen later
	M1	For multiplying numerator **and** denominator by their conjugate which must of the form e.g $\sqrt{A}\pm\sqrt{B}$ **AND** simplifying to the required form. We need to see at least one intermediate line of working before the final answer here.
	A1 cso	The answer as shown with no errors or omissions seen. This is a given result.
(c)	M1	For finding the length FX using any appropriate trigonometry.
	M1	For using any appropriate trigonometry to find length EF [or XY]
	A1	For awrt 7.1 (cm)
(d)	B1	For identifying the correct angle required.
	M1	For using Pythagoras theorem to find FC $FC=\sqrt{21.51781^2+7.09505^2}=22.65735.....$ Or For using Pythagoras theorem to find CX $FC=\sqrt{15.213^2+7.095^2}=16.786...$ [Note: $BX=15.213...$ and $CX=16.786\ldots$]
	M1	For using any appropriate trigonometry to find angle FCX. For example; $\angle FCX=\tan^{-1}\left(\frac{15.2153}{16.786}\right)=42.186...^\circ\approx42.2^\circ$
	A1	For awrt 42.2°

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USEFUL SKETCH



NB:

$AC = 25.03$ cm

$AF = 17.57$ cm

$CF = 22.65$ cm

Angle $FCA = 42.9°$

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Question	Scheme	Marks

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Answer 5 of 5 · Question 11 part b · 2 marks

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11(a)	Let the intersection of AC and DB be X $[AC =]\sqrt{6^2+6^2}\left[=\sqrt{72} \Rightarrow AX = 3\sqrt{2}\right]$ oe or $[AX =]\sqrt{3^2+3^2}\left[=\sqrt{18}=3\sqrt{2}\right]$ $[h =]\sqrt{12^2-"\left(3\sqrt{2}\right)"^2} = 3\sqrt{14}$ oe*	M1 M1A1 cso [3]
ALT1	Let Z be the midpoint of AB $[EZ =]\sqrt{12^2-3^2}\left[=\sqrt{135}=3\sqrt{15}\right]$ oe $[h =]\sqrt{\left("3\sqrt{15}"\right)^2-(3)^2} = 3\sqrt{14}$ oe*	M1 M1A1 cso [3]
ALT2	$[h =]\sqrt{3^2+3^2+h^2} = 12$ or $3^2+3^2+h^2 = 144$	M1M1 A1
(b)	Required angle $= [\angle EAX]\sin^{-1}\left(\frac{3\sqrt{14}}{12}\right)$ or $\tan^{-1}\left(\frac{3\sqrt{14}}{3\sqrt{2}}\right)$ or $\cos^{-1}\left(\frac{3\sqrt{2}}{12}\right) = 69.295...^\circ$ or $\cos^{-1}\left(\frac{\left(6\sqrt{2}\right)^2+12^2-12^2}{2\times 6\sqrt{2}\times 12}\right) \approx 69.3^\circ$	M1A1 [2]

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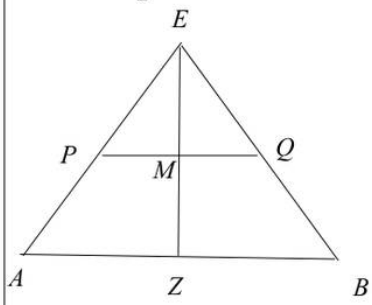
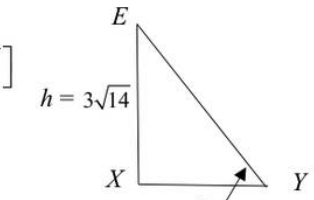
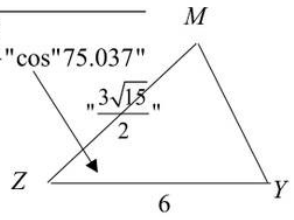
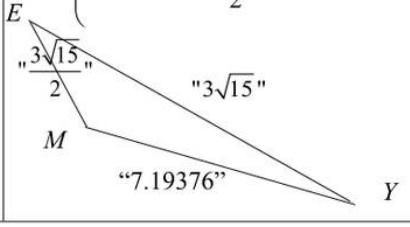
2025 NOV · Paper 02 · extract 2 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For correctly applying Pythagoras theorem to find AC or AX Full method, must include square rooting.
	M1	For correctly applying Pythagoras theorem to find h Allow follow through of their AX/AC. Full method, must include square rooting. Not a dependent method mark, but must see M1 or be correctly using $3\sqrt{2}$ M0 M1 A0 will occur. We will permit eg $3\sqrt{2}^2$ if the intention is clear
	A1*cso	For the correct length only. Minimum steps shown, no errors or omissions.
ALT1	M1	For correct applying Pythagoras theorem to find EZ Full method, must include square rooting.
	M1	For correctly applying Pythagoras theorem to find h Allow follow through of their EZ. Full method, must include square rooting. Not a dependent method mark, but must see M1 or be correctly using $3\sqrt{15}$ M0 M1 A0 will occur. We will permit eg $3\sqrt{15}^2$ if the intention is clear
	A1*cso	For the correct length only. Minimum steps shown, no errors or omissions.
ALT2	M1M1	For a fully correct statement using 3D Pythagoras, each side shown. M0 M0 for any part wrong
	A1*cso	For the correct length only. Minimum steps shown, no errors or omissions.
(b)	M1	For using the given value of h and applying correct trigonometry to find the required angle. Allow ft of $3\sqrt{2}$ if from correct method. If any other method used, the **method** must be complete and correct ie leaving as sin = is not complete. Note equivalent angles (because of symmetry) eg $\angle EBD$
	A1	For the correct angle awrt 69.3°

Type MS · Angle between a line and a plane

Answer 5 of 5 · Question 11 part b · 2 marks

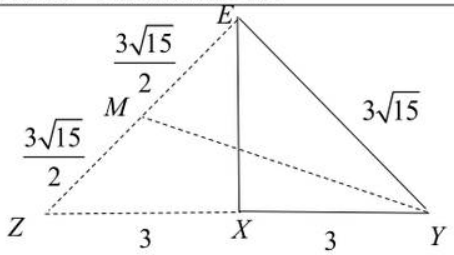
2025 NOV · Paper 02 · extract 3 of 5 · find part b in this mark scheme extract

(c)	Let M be the midpoint of PQ, Y be the midpoint of DC and Z be the midpoint of AB	
	$[EY = EZ =]\sqrt{3^2 + \left(3\sqrt{14}\right)^2}\ [= 3\sqrt{15}]$	M1
	$[EM =]\frac{"3\sqrt{15}"}{2}$	M1
	 	
	Angle between planes EAB & ABC =	
	$\left[[\angle EYX =]\tan^{-1}\left(\frac{3\sqrt{14}}{3}\right)\left[= 75.0367^\circ\right]\right.$ $\left.\text{or } \cos^{-1}\left(\frac{"\left(3\sqrt{15}\right)"^2 + 6^2 - "\left(3\sqrt{15}\right)"^2}{2\times"3\sqrt{15}"\times 6}\right) \text{ or } \cos^{-1}\left(\frac{3}{"3\sqrt{15}"}\right) \text{ or } \sin^{-1}\left(\frac{3\sqrt{14}}{"3\sqrt{15}"}\right)\right]$	M1
	$[MY =]\sqrt{6^2 + \left("\frac{3\sqrt{15}}{2}"\right)^2 - 2\times 6\times"\frac{3\sqrt{15}}{2}"\cos"75.037"}$ $= 7.19376...$ 	A1 (M1 on ePen)
	[Required angle = $\angle EMY$ =]	
	$\cos^{-1}\left(\frac{\left("\frac{3\sqrt{15}}{2}"\right)^2 + "7.19376"^2 - \left("3\sqrt{15}"\right)^2}{2\times"\frac{3\sqrt{15}}{2}"\times"7.19376"}\right) = 126.3133 = 126$	dddM1 A1
		[6]

Type MS · Angle between a line and a plane

Answer 5 of 5 · Question 11 part b · 2 marks

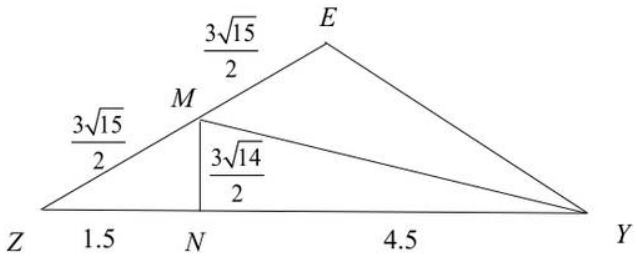
2025 NOV · Paper 02 · extract 4 of 5 · find part b in this mark scheme extract

3rd M mark and 1st A mark – ALTs to find MY		
	 $\left[\left[\angle ZEY = 2\times\angle XEY =\right] 2\times\tan^{-1}\left(\frac{3}{3\sqrt{15}}\right)\left[= 2\times 14.9632^{\circ}\right]\right]$ or $\left[\left[\angle ZEY =\right]\cos^{-1}\frac{"\left(3\sqrt{15}\right)"^2+"\left(3\sqrt{15}\right)"^2-"(6)"^2}{2\times"3\sqrt{15}"\times"\left(3\sqrt{15}\right)"}\left[= 29.9264\right]\right]$	M1
	$MY = \sqrt{\left("3\sqrt{15}"\right)^2+\left("\frac{3\sqrt{15}}{2}"\right)^2-2\times"3\sqrt{15}"\times"\frac{3\sqrt{15}}{2}"\cos 29.9264}$ $= 7.19376...$	A1
		Total 11 marks

Type MS · Angle between a line and a plane

Answer 5 of 5 · Question 11 part b · 2 marks

2025 NOV · Paper 02 · extract 5 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks
11c ALT	Dropping a perpendicular from M to the base of the pyramid, which hits at point N  $(MN=)\frac{3\sqrt{14}}{2}$ $\left(\text{Note: } MZ=\sqrt{\left(\frac{3\sqrt{14}}{2}\right)+1.5^2}=\frac{3\sqrt{15}}{2}\right)$ $(\angle ZMN=)\tan^{-1}\left(\frac{1.5}{\frac{3\sqrt{14}}{2}}\right)$ or $\cos^{-1}\left(\frac{\frac{3\sqrt{14}}{2}}{\frac{3\sqrt{15}}{2}}\right)$ or $\sin^{-1}\left(\frac{1.5}{\frac{3\sqrt{15}}{2}}\right)(=14.96321743)$ $(\angle YMN=)\tan^{-1}\left(\frac{4.5}{\frac{3\sqrt{14}}{2}}\right)(=38.7220711)$ $(\angle YMZ=14.96321743+38.7220711=)53.68528853$ $180-"53.68528853"=126.3$	M1 M1 M1 A1 dddM1A1 [6]

Part	Mark	Notes
ALT	M1	For correctly stating (explicitly or in working) the height of MN
	M1	For the full and correct **method** to find $\angle ZMN$ They may follow through MZ if found and from a correct method.
	M1	For the full and correct **method** to find $\angle YMN$ They may follow through MZ if found and from a correct method.
	A1	For the correct angle $\angle YMZ$ allow awrt 39
	dddM1	Dependent on all previous method marks, for subtracting their angle from 180.
	A1	For awrt 126°